

CODED — Read & Preserve Earth Data on IPFS

Open a real **2 GB Icechunk dataset** straight from a content ID — then make your own Zarr or Icechunk data takedown-resistant with one upload. (Zarr · Icechunk · xarray · Storacha/Filecoin)

Cory Levinson · Rich Signell · Brian Terry

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Read it live — by its CID

This **runs today** — no daemon, no credentials. Just `pip install` and open a 2 GB content-addressed ERA5 dataset over HTTP; Icechunk 2.0 verifies every byte against the hash.

In [1]:

```
import icechunk, xarray as xr

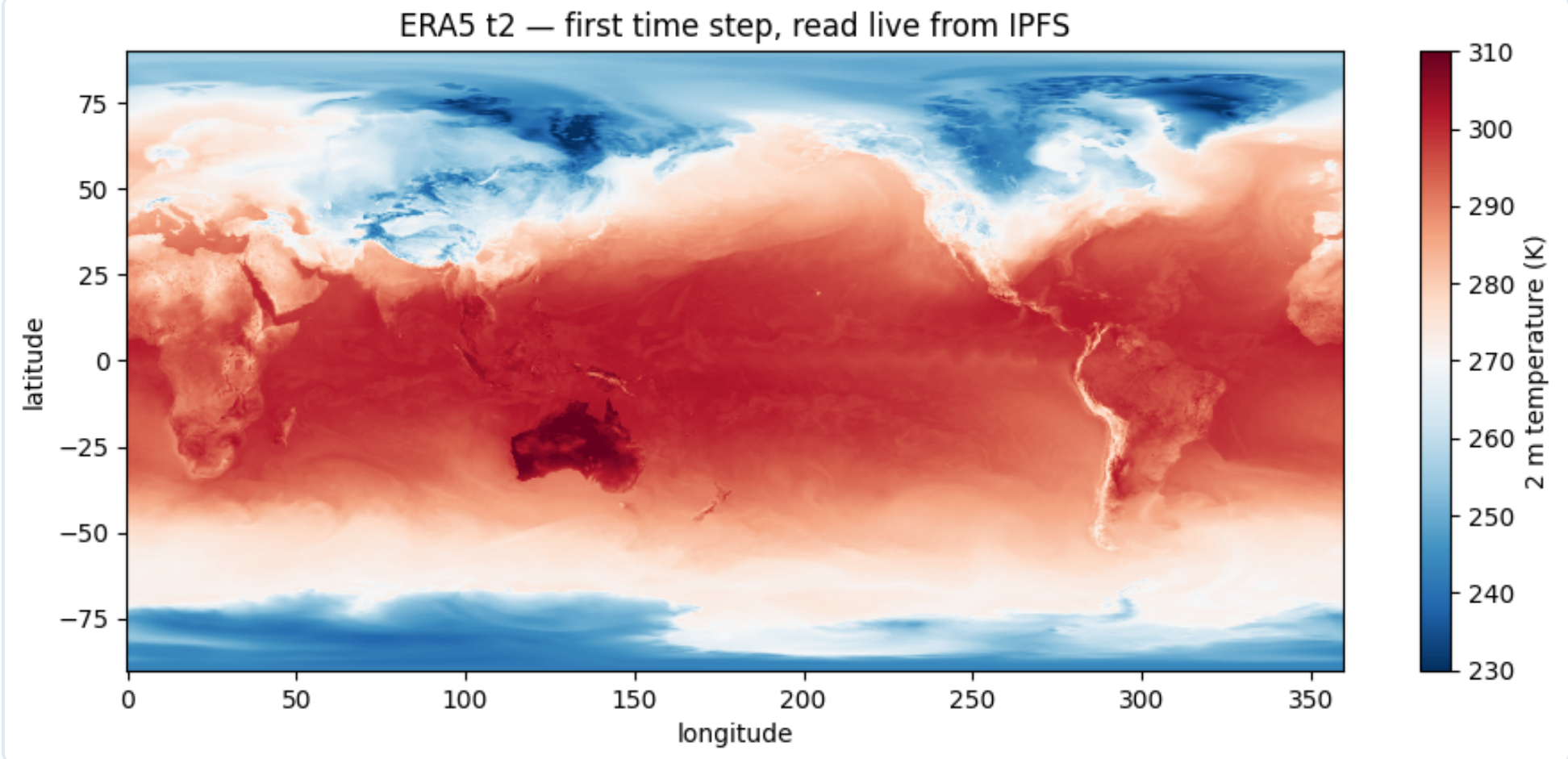
CID = "bafybeicwoya7...avptj7y"
# try the CODED gateway; fall back to ipfs.io
GATEWAYS = ["http://44.239.118.35:8080",
            "https://ipfs.io"]

for GATEWAY in GATEWAYS:
    try:
        st = icechunk.http_storage(f"{GATEWAY}/ipfs/{CID}")
        repo = icechunk.Repository.open(st)
        ds = xr.open_zarr(repo.readonly_session("main").store,
                        chunks={})
        print(f"✓ opened via {GATEWAY}"); break
    except Exception as e:
        print(f"✗ {GATEWAY} failed; trying next...")
```

✓ opened via http://44.239.118.35:8080

In [2]:

```
ds.t2.isel(time=0).plot(cmap="RdBu_r") # one field, fetched lazily
```



ERA5 2-m temperature (K), first time step — read live from IPFS by CID.

area-weighted global-mean 2 m temperature = 286.5062 K
read 2076 MB in 15.5 s over IPFS
✓ bit-for-bit match with the reference value (286.5062 K)

Two gateways, plain `try/except`: the fast co-located node serves it in **0.6 s**; if that node is gone, it falls straight through to the public `ipfs.io`. Same CID, same verified bytes, everywhere.

What just happened

A **2 GB, versioned, content-addressed** dataset, opened lazily over plain HTTP — no daemon, no credentials.

0.6 s
open the repo
(metadata only)

15.5 s
stream full 2 GB
global mean

100%
byte-verified
by the CID

Preserve your dataset

Have a Zarr or Icechunk dataset you can't afford to lose? Write it to **Storacha** (Filecoin-backed, free to 5 GB) and get a permanent **CID** anyone can verify — independent of any single institution.

- 1

Build / point at your **Icechunk** store
`ds.to_zarr(icechunk_store)`
- 2

Add to IPFS → content ID
`ipfs add -r --cid-version=1 store/`
- 3

Pin to Storacha/Filecoin
`w3 up --car store.car`

```
# one command with our coded-pin tool:
coded-pin native my.zarr \
  --name my-dataset \
  --pin storacha \
  --ipns my-dataset-current
# → CID bafybei... (share it alongside your DOI)
```

The **CID is a hash of the data** — tamper-evident and location-independent. Add an **IPNS** name for a mutable pointer that follows every new version.

Why this matters

- **Content-addressed.** The CID is the data's fingerprint — verify you have exactly the right bytes, from anywhere.
- **Takedown-resistant.** One budget cut or policy change can't erase data that many nodes can pin and serve.
- **Zero adapter code.** Icechunk's `http_storage` reads a repo over any IPFS gateway — standard xarray/Zarr the whole way.
- **Reproducible.** Every Icechunk commit is an immutable snapshot; an old CID always returns the old data.

Bottom Line

Reading distributed Earth data is a one-cell, no-credentials operation today — open any Icechunk/Zarr dataset by its CID, with automatic gateway fallback.

Preserving it is just as easy: one upload to Storacha mints a verifiable, Filecoin-backed CID. Publish it next to your DOI and your data outlives any single server.